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Semantic Segmentation on 3D Occupancy Grids for Automotive Radar

ROBERT PROPHET[1], (Student Member, IEEE), ANASTASIOS DELIGIANNIS[2], (Member, IEEE), JUAN-CARLOS FUENTES-MICHEL[2], INGO WEBER[2], AND MARTIN VOSSIEK[1], (Fellow, IEEE)

[1]Institute of Microwaves and Photonics, Friedrich-Alexander-University Erlangen-Nuremberg, 91058 Erlangen, Germany
[2]BMW Group, 80788 Munich, Germany

Corresponding author: Robert Prophet (robert.prophet@fau.de)

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ABSTRACT Radar sensors have great advantages over other sensors in estimating the motion states of moving objects, because they detect velocity components within one measurement cycle. Moreover, numerous successes have already been achieved regarding the classification of such objects. However, the advantage of instantaneous velocity measurement is lost when detecting static objects, so that their classification is much more demanding. In this paper, we use semantic segmentation networks to distinguish between frequently occurring infrastructure objects. The resulting semantic grids provide a location-based classification of the vehicle environment. Since even modern radars have a significantly poorer angular resolution than lidars, the relatively thin radar point cloud is accumulated in advance and transformed into 2D or 3D grids that act as network inputs. Occupancy grids are particularly advantageous here, since they calculate not only the obstacles but also the free spaces. With suitable parameter selection, which is very challenging due to the complexity of radar measurement, the resulting grids allow for good association with camera images. Finally, in order to evaluate possible advantages of 3D grids as network input with respect to the segmentation result, we created and evaluated a simulation dataset and two different real-world datasets in car parks and on motorways. As a result, Jaccard coefficients between 81% and 88% were achieved, depending on the dataset. It was also found that 3D input images lead to improvements in the car park dataset.

INDEX TERMS 79 GHz, automotive radar, deep learning, occupancy grid, semantic segmentation.

I. INTRODUCTION

Ultrasonic, camera, lidar and radar sensors are used to detect the surroundings of autonomous vehicles. Radar sensors are characterized by their robustness against changing weather conditions and for the instantaneous measurement of the Doppler velocity. A disadvantage is the comparatively poor angular resolution.

Multiple recent publications have dealt with improvements in object-based, dynamic environment models. This mainly includes the grouping of radar targets to the corresponding object [1], the estimation of object geometry and state of motion [2], and finally their assignment to an object class, such as pedestrian or car [3]–[5]. For fully autonomous driving, however, the creation of a static environment model is also indispensable, for example, to identify the road course ahead according to the infrastructure. Here, map-based environment models are preferred because they can describe the current free space geometrically. These environment models are also called grids as the entire observation area is discretized in 2D or 3D cells (pixels or voxels).

Given that even modern automotive radars measure relatively few targets in a measurement cycle, as shown in Fig. 1, the static targets are accumulated over time to generate a detailed environment model. A Cartesian local grid's targets are continuously integrated into a global grid, whose *xy*-plane is usually parallel to the ground. This accumulation results in a much more detailed model of the environment, which allows a good association with a camera image taken from a bird's eye view. The individual cells contain either information about the radar cross section (RCS) or the occupancy status, so that one speaks of RCS grids (RG) and occupancy grids (OG).

In addition to these two types, there are so-called semantic grids (SG), which assign a specific object class to each cell. Knowledge of the object class is essential for driver assistance

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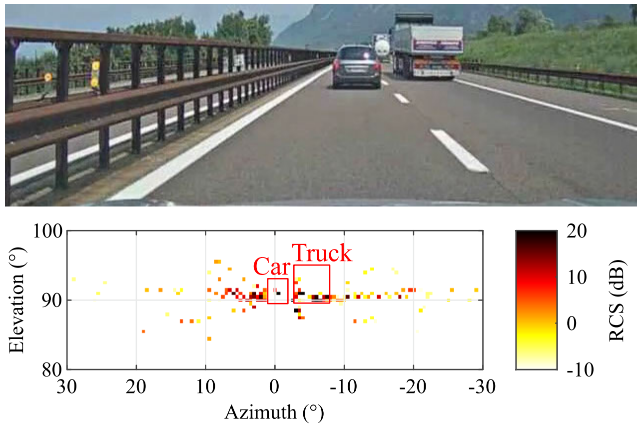

FIGURE 1. Comparison of camera and front radar images. Each dot in the radar image shows the largest measured radar cross section (RCS) for each solid angle, which is a measure of detectability. Obviously, the radar image is much sparser than the camera image.

systems, since, for example, cars parked during the parking process are treated differently from fences. The authors of [6] show one of the first attempts to create such SGs. Areas of interest within the SGs were extracted with a clustering algorithm and then investigated with analytical methods. On the other hand, [7] uses convolutional neural networks (CNN) to investigate the image details. Finally, [4] and [8] achieved good results with complete pixel-wise classifications using extensive CNNs, thus avoiding the error-prone clustering step.

All existing publications have in common that classification is based on 2D grids. Therefore, in this paper, we want to use 3D radar target lists and convert them into 3D grids, which are then transformed to SGs via CNN in order to compare the results with those of the 2D input grids. Fig. 2 shows schematically the individual steps of the entire processing chain including a brief explanation. Clearly, different objects provide different 3D reflection patterns, which would not be exploited in a 2D approach.

The paper is structured as follows: After a detailed description of the generation of 2D and 3D grids under consideration of radar specifications in Section II, the CNNs used are presented in Section III. Afterwards, in Section IV we present the datasets used to evaluate the process and explain the labeling procedure, which is much less intuitive than for camera images. Finally, Section V presents the results obtained before Section VI summarizes the findings.

II. GRID PROCESSING

A. POINT CLOUD GENERATION

The starting point of our grid processing system are the target lists of the K radar sensors of the test vehicle in the current measurement cycle t. These target lists are first transformed into a current point cloud $act(t)$, see Fig. 2 (b), and an accumulated point cloud $acc(t)$. If the k-th sensor, $k \in \{1, \ldots, K\}$, measures $N_k(t)$ targets in the measurement cycle t, the range $R_{i,k}^{(\mathrm{sc}_k)}$, the relative radial velocity $v_{i,k}^{(\mathrm{sc}_k)}$, the azimuth $\varphi_{i,k}^{(\mathrm{sc}_k)}$, the elevation $\theta_{i,k}^{(\mathrm{sc}_k)}$ and the RCS $\sigma_{i,k}^{(\mathrm{sc}_k)}$ of each target $i \in I_k(t) = \{1, \ldots, N_k(t)\}$ are obtained in the sensor coordinate system sc_k. Since modern automotive radars use fast chirp modulation [9], $v_{i,k}^{(\mathrm{sc}_k)}$ is only measured in a specific unambiguity range v_{unamb}. If N_{chirp} and T_{seq} denote the number of chirps and the sequence duration (consecutive chirps processed coherently) and λ_{c} denotes the carrier wavelength, then according to Shannon's sampling theorem it follows

$$v_{\mathrm{unamb}} = \pm \frac{N_{\mathrm{chirp}}}{4} \cdot \frac{\lambda_{\mathrm{c}}}{T_{\mathrm{seq}}}. \tag{1}$$

Accordingly, the shorter the time between two consecutive chirps, the greater v_{unamb}. Especially in highway scenarios, this value is often too low even with modern sensors, since relative speeds of over 200 km/h are regularly encountered. To address this challenge, we use the algorithm in [10] to extend v_{unamb}: If two consecutive sequences or measurement cycles are measured with the unambiguous ranges $v_{\mathrm{unamb},1}$ and $v_{\mathrm{unamb},2}$, the velocity of the target can be calculated according to the Chinese remainder theorem in a significantly larger range equal to the least common multiple of $v_{\mathrm{unamb},1}$ and $v_{\mathrm{unamb},2}$. However, the targets from the two measurement cycles need to be correctly assigned for this approach to work. We use a joint probabilistic data association filter [11] for target assignment here. Furthermore, the actual relative radial velocities of the targets must remain approximately constant during the two measurement cycles.

Since the objects to be classified are static, all moving radar targets (hereinafter: dynamic) must be identified in order to filter them out later. This is classically done by comparing $v_{i,k}^{(\mathrm{sc}_k)}$ with the radial component of the k-th sensor speed, which is why it is critical that v_{unamb} is extended. The sensor speed can be calculated from the test vehicle's odometric data. If said radial component deviates too much from $v_{i,k}^{(\mathrm{sc}_k)}$, the target is dynamic and has the index $i \in D_k(t) \subseteq I_k(t)$. Thus $D_k(t)$ is the set of all dynamic targets indices of the k-th sensor in the current measurement cycle and $S_k(t) = I_k(t) \backslash D_k(t)$ is the corresponding set of all static targets.

The point cloud $act\,(t) = act_{\mathrm{stat}}(t) \cup act_{\mathrm{dyn}}(t)$ to be generated now represents all static $act_{\mathrm{stat}}(t)$ and dynamic $act_{\mathrm{dyn}}(t)$ radar targets in the measurement cycle t. For the representation we use on the one hand a local, Cartesian 3D coordinate system lc, whose xy-plane is parallel to the ground with the x-axis oriented in the forward direction with respect to the current test vehicle position. The coordinate origin is the vertical projection of the center of the rear axle onto the ground. On the other hand, we use a global, Cartesian 3D coordinate system gc, which corresponds to the local coordinate system of the very first measurement cycle. Since the length of our grids does not exceed 1 km in any dimension, the curvature of the earth's surface can be neglected and flat ground can be assumed with very good approximation. The targets are transformed from the sensor coordinate systems into lc with the transformation functions $T_k = T_k\left(R_{i,k}^{(\mathrm{sc}_k)}, \varphi_{i,k}^{(\mathrm{sc}_k)}, \theta_{i,k}^{(\mathrm{sc}_k)}\right)$, which, with knowledge of the individual sensor mounting positions, perform the corresponding rotations and

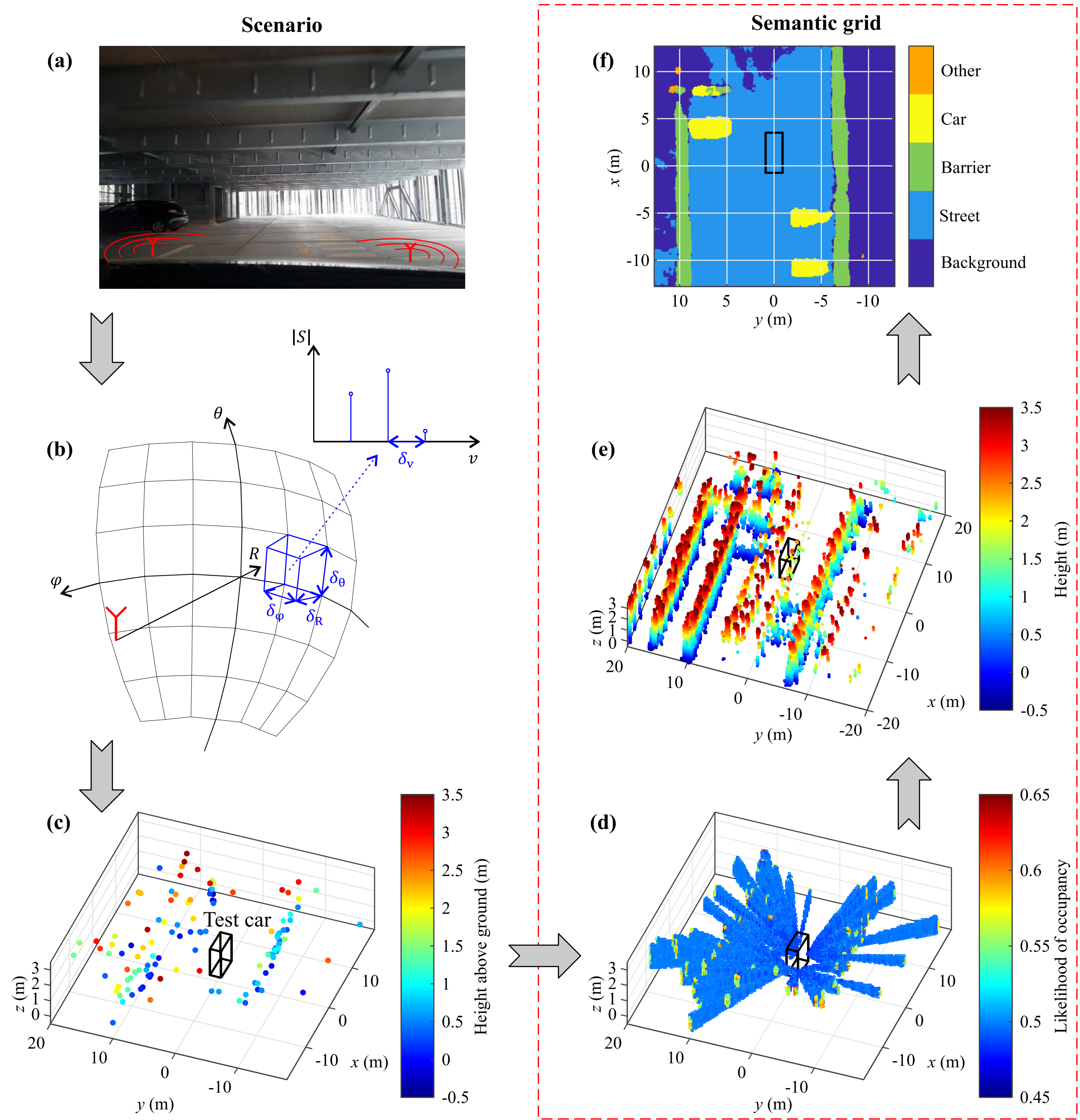

FIGURE 2. **Processing chain. Scenario (a) is scanned with the test vehicle radars, where each sensor resolves the spectral amplitude S in azimuth φ, elevation θ, range R and radial velocity v with δ_φ, δ_θ, δ_R and δ_v, compare (b). The result of the basic processing step is one target list per measurement cycle, which represents all radar targets of all sensors in a local vehicle coordinate system (c). The black cuboid symbolizes the test vehicle here. This is followed by the processing discussed in this article and framed in red in the figure. First the target list is transferred to a local OG (d), which is then integrated into a global OG (e). The global OG image quality is significantly enhanced by a process of continuous accumulation, whereby the target lists from all measurement cycles are integrated in the OG. The figure shown here shows all voxels that exceed a certain occupancy probability. Finally, the global OG acts as the input to a CNN, whose output is the SG (f).**

translations to the lc. With respect to $act_{\mathrm{stat}}(t)$, the Cartesian coordinates of the static targets $\boldsymbol{x}^{(\mathrm{lc})}_{s_{\mathrm{act}}} = \left(x^{(\mathrm{lc})}_{s_{\mathrm{act}}}, y^{(\mathrm{lc})}_{s_{\mathrm{act}}}, z^{(\mathrm{lc})}_{s_{\mathrm{act}}}\right)^T$ are obtained with $s_{\mathrm{act}} \in S_{\mathrm{act}}(t) = \{1, \ldots, \sum_{k=1}^{K} |S_k(t)|\}$ and with respect to $act_{\mathrm{dyn}}(t)$ the dynamic targets $\boldsymbol{x}^{(\mathrm{lc})}_{d_{\mathrm{act}}}$ with $d_{\mathrm{act}} \in D_{\mathrm{act}}(t) = \{1, \ldots, \sum_{k=1}^{K} |D_k(t)|\}$.

The accumulated point cloud $acc(t)$ contains all static coordinates $\boldsymbol{x}^{(\mathrm{gc})}_{s_{\mathrm{acc}}}$ of all completed measurement cycles $1 \ldots t$ with $s_{\mathrm{acc}} \in S_{\mathrm{acc}}(t) = \{1, \ldots, N_{\mathrm{acc}}(t)\}$. $N_{\mathrm{acc}}(t)$ is the number of all radar targets in $acc(t)$. Thus, the accumulation consists of a measurement cycle-wise integration of the targets from $act_{\mathrm{stat}}(t)$ into $acc(t)$. To carry out the accumulation successfully, the current positional relationship between lc and gc must be known. This is obtained by continuously calculating the test vehicle's pose between two successive measurement cycles. Because vertical movements can only be roughly calculated with onboard sensors, we assume a 2D pose $\boldsymbol{X}_{\text{ego}}(t) = \left(x_{\text{ego}}^{(\text{gc})}(t), y_{\text{ego}}^{(\text{gc})}(t), \alpha_{\text{ego}}^{(\text{gc})}(t)\right)^{\text{T}}$. This describes the current position in the xy-plane and the course angle, and its change $\Delta\boldsymbol{X}_{\text{ego}}(t)$ from measurement cycle $t-1$ to t. We realize the calculation by applying the constant turn rate and acceleration model [12] based on odometric data, and a radar-specific simultaneous localization and mapping algorithm [13]. Global navigation satellite systems are not used.

All road users have mass elements, such as, those parts of a car's tires that are currently in contact with the ground, whose instantaneous speed is almost zero relative to the ground, even during locomotion. For this reason, radars regularly detect static targets for dynamic objects. Furthermore, objects that were once at rest can start to move, so that the static targets in $acc\,(t)$ cannot be assigned to any real object. To correct $acc\,(t)$ for this, we remove all targets with the indices from the set $Del(t) \subseteq S_{\text{acc}}(t-1)$ if their distance to at least one target in $act_{\text{dyn}}(t-1)$ is less than ε_{dyn}:

$$Del(t) = \Big\{s_{\text{acc}} \in S_{\text{acc}}(t-1) \big| |\boldsymbol{x}_{s_{\text{acc}}}^{(\text{gc})} - \boldsymbol{x}_{d_{\text{act}}}^{(\text{gc})}| < \varepsilon_{\text{dyn}} \forall d_{\text{act}} \in D_{\text{act}}(t-1)\Big\} \tag{2}$$

The accumulated point cloud $acc(t)$ is thus continuously corrected in every measurement cycle and finally consists of the targets with the indices from the set $S_{\text{cor}}(t) = S_{\text{acc}}(t) \backslash Del(t)$.

In summary, the point clouds are described by the following equations:

$$act_{\text{stat}}(t) = \left\{\left(\boldsymbol{x}_{s_{\text{act}}}^{(\text{gc})}, \boldsymbol{f}_{s_{\text{act}}}\right) \mid \boldsymbol{x}_{s_{\text{act}}}^{(\text{gc})} \in \mathbb{R}^3, \boldsymbol{f}_{s_{\text{act}}} \in \mathbb{R}^{N_{\text{f}}}, s_{\text{act}} \in S_{\text{act}}(t)\right\} \tag{3}$$

$$act_{\text{dyn}}(t) = \left\{\left(\boldsymbol{x}_{d_{\text{act}}}^{(\text{gc})}, \boldsymbol{f}_{d_{\text{act}}}\right) \mid \boldsymbol{x}_{d_{\text{act}}}^{(\text{gc})} \in \mathbb{R}^3, \boldsymbol{f}_{d_{\text{act}}} \in \mathbb{R}^{N_{\text{f}}}, d_{\text{act}} \in D_{\text{act}}(t)\right\} \tag{4}$$

$$acc(t) = \left\{\left(\boldsymbol{x}_{s_{\text{acc}}}^{(\text{gc})} \boldsymbol{f}_{s_{\text{acc}}}\right) \mid \boldsymbol{x}_{S_{\text{acc}}}^{(\text{gc})} \in \mathbb{R}^3, \boldsymbol{f}_{s_{\text{acc}}} \in \mathbb{R}^{N_{\text{f}}}, S_{\text{acc}} \in S_{\text{cor}}(t)\right\} \tag{5}$$

Here, $\boldsymbol{f}$ denotes an N_{f}-dimensional property vector which contains, among other things, the RCS and certain measurement uncertainties. Fig. 3 illustrates the most important relationships in this section.

B. OCCUPANCY GRID

1) BASICS

As already mentioned in Section I, the OG is a map-based environment model that was presented in detail in [14]. The environment is divided into cells of equal size with edge length l_{og}, where the n-th cell has the occupancy state $m_n^{(\text{og})}(t)$ in the measurement cycle t. The totality of all occupancy states is defined with $\boldsymbol{m}^{(\text{og})}(t)$. The associated coordinate system always corresponds to gc.

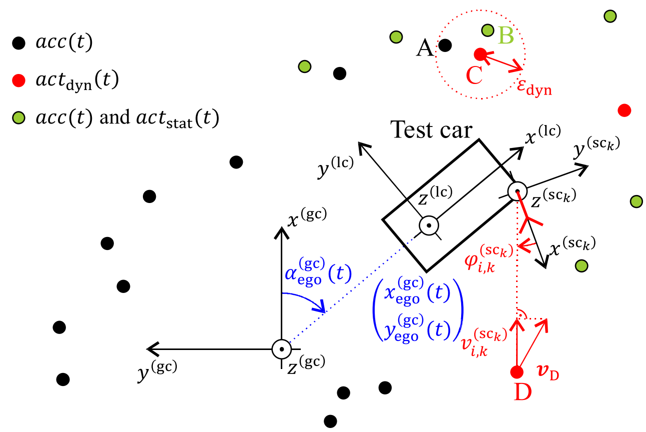

FIGURE 3. Overview of the defined coordinate systems and point clouds. The targets A and B are removed from $acc(t+1)$ because they are close enough to C. The radar sensor measures only the radial component $v_{i,k}^{(\text{sc}_k)}$ of the relative velocity v_D, which is a function of azimuth $\varphi_{i,k}^{(\text{sc}_k)}$ and elevation $\theta_{i,k}^{(\text{sc}_k)}$.

The occupancy state is a binary random variable which can take the values 1 for occupied or 0 for free. The corresponding posterior probability is $p\left(m_n^{(\text{og})}(t)\right)$. Assuming independent cells, the posterior probability of the entire OG is

$$p\left(\boldsymbol{m}^{(\text{og})}(t) \mid act_{\text{stat}}(1\ldots t), \boldsymbol{X}_{\text{ego}}(1\ldots t)\right) = \prod_n p\left(m_n^{(\text{og})}(t) \mid act_{\text{stat}}(1\ldots t), \boldsymbol{X}_{\text{ego}}(1\ldots t)\right) \tag{6}$$

Consequently, the OG uses the information from the current static point clouds from all completed measurement cycles as well as the test vehicle positions at the time of the measurements. Under the given assumptions, (6) can be calculated using a Bayesian filter and, using logits, $l\left(m_n^{(\text{og})}\right) = \log\left(p\left(m_n^{(\text{og})}\right) \Big/ \left(1 - p(m_n^{(\text{og})})\right)\right)$ for each cell affected by the measurement:

$$\begin{aligned} l\left(m_n^{(\text{og})}(t) \mid act_{\text{stat}}(1\ldots t), \boldsymbol{X}_{\text{ego}}(1\ldots t)\right) &= l\left(m_n^{(\text{og})}(t) \mid act_{\text{stat}}(t), \boldsymbol{X}_{\text{ego}}(t)\right) \\ &\quad + l\left(m_n^{(\text{og})}(t-1) \mid act_{\text{stat}}(1\ldots t-1), \boldsymbol{X}_{\text{ego}}(1\ldots t-1)\right) \\ &\quad - l\left(m_n^{(\text{og})}(1)\right). \end{aligned} \tag{7}$$

Here, the last term on the right represents the logit of the prior probability, which is usually zero in this context, and the second term represents the recursive part. The first term describes the so-called inverse sensor model (ISM), which defines the modelling of targets (positive ISM) and free spaces (negative ISM) depending on the dimension of the OG.

Furthermore, we avoid overly strong accumulation effects by using a maximum and minimum logit $l_{\max}$ and $l_{\min}$, which are applied after integrating all targets in measurement cycle t

into the OG:

$$l\left(m_n^{(\mathrm{og})}(t)\right)=\begin{cases} l_{\max} \forall l\left(m_n^{(\mathrm{og})}(t)\right)>l_{\max} \\ l_{\min} \forall l\left(m_n^{(\mathrm{og})}(t)\right)<l_{\min}. \end{cases} \tag{8}$$

2) POSITIVE INVERSE SENSOR MODEL IN 3D

To calculate the occupation probability of voxels in the vicinity of the i-th radar target we use the probability density function (PDF) $P_i^{(\mathrm{sc}_k)}$, $i \in S_{\mathrm{act}}(t)$ (the index s_{act} is omitted in this section for better readability), which describes a Gaussian function in the sensor coordinate system sc_k associated with i:

$$\begin{aligned} &P_i^{(\mathrm{sc}_k)}(R,\varphi,\theta) \\ &= A_i \exp\left(-\left(\frac{(R-R_i)^2}{2SD_{\mathrm{R},i}^2}+\frac{(\varphi-\varphi_i)^2}{2SD_{\varphi,i}^2}+\frac{(\theta-\theta_i)^2}{2SD_{\theta,i}^2}\right)\right) \end{aligned} \tag{9}$$

The corresponding standard deviations are $SD_{\mathrm{R}}=SD_{\mathrm{R}}(\sigma)$, $SD_{\varphi}=SD_{\varphi}(\sigma,\varphi,\theta)$ as well as $SD_{\theta}=SD_{\theta}(\sigma,\theta)$ and A is the maximum value of the PDF. According to [15], the dependence of the standard deviations on the RCS σ results from the relationship between standard deviation SD, resolution δ and the signal-to-noise ratio $SNR \propto \sigma$:

$$SD \propto \frac{\delta}{\sqrt{2SNR}}. \tag{10}$$

Relationship (10) specifies a lower limit of accuracy, which in this case applies to all three standard deviations. The standard deviations in the angular dimensions also depend on the angles themselves, since half the main lobe width of the array factor in azimuth $\delta_{\varphi,i}$ and elevation $\delta_{\theta,i}$ for a planar antenna array with aperture lengths D_{y} and D_{z} becomes larger the more the direction of the target (φ_i,θ_i) deviates from the antenna orientation $(0°, 90°)$ [16]:

$$\begin{aligned} \delta_{\varphi,i} \cong{}& \cos^{-1}\left((\sin\varphi_i\cdot\sin\theta_i-0.45\frac{\lambda_{\mathrm{c}}}{D_{\mathrm{y}}})/\sin\theta_i\right) \\ &-\cos^{-1}\left((\sin\varphi_i\cdot\sin\theta_i+0.45\frac{\lambda_{\mathrm{c}}}{D_{\mathrm{y}}})/\sin\theta_i\right), \end{aligned} \tag{11}$$

$$\begin{aligned} \delta_{\theta,i} \cong{}& \cos^{-1}\left(\cos\theta_i-0.45\frac{\lambda_{\mathrm{c}}}{D_{\mathrm{z}}}\right) \\ &-\cos^{-1}\left(\cos\theta_i+0.45\frac{\lambda_{\mathrm{c}}}{D_{\mathrm{z}}}\right). \end{aligned} \tag{12}$$

The 4D hypervolume V_i within the volume $P_i^{(\mathrm{sc}_k)}$ is expressed as

$$\begin{aligned} V_i &= \iiint_{-\infty}^{\infty} P_i^{(\mathrm{sc}_k)} dR\,d\varphi\,d\theta \\ &= (2\pi)^{\frac{3}{2}} A_i\cdot SD_{\mathrm{R},i}SD_{\varphi,i}SD_{\theta,i}. \end{aligned} \tag{13}$$

Since the probability of a false alarm depends only on the RCS, V_i should also depend only on the RCS, analogous to [17], so that we calculate the maximum value in (9) as follows:

$$A_i=\frac{A_{\max}(\sigma_i)}{\frac{SD_{\mathrm{R},i}}{SD_{\mathrm{R},\min}}\cdot\frac{SD_{\varphi,i}}{SD_{\varphi,\min}}\cdot\frac{SD_{\theta,i}}{SD_{\theta,\min}}}. \tag{14}$$

Terms $SD_{\mathrm{R},\min}$, $SD_{\varphi,\min}$ and $SD_{\theta,\min}$ are the constant standard deviations in the case of the highest possible plausibility of the measurement, in which A_i then takes the value $A_{\max}(\sigma_i)$.

Since the radar sensor measures in spherical coordinates, $P_i^{(\mathrm{sc}_k)}$ is dependent on R_i in the lateral directions after transformation into a Cartesian coordinate system. The volumes appear expanded with increasing distance, which has a negative effect on the image quality of the OG, especially for radars with large ranges. To counteract this, $A_{\max}$ in [14] can be replaced by an A_{cor}, which, following the example of [18], has an additional distance dependence, which is controlled by the scaling factor R_{scal}:

$$A_{\mathrm{cor}}(\sigma_i,R_i)=A_{\max}(\sigma_i)\cdot\exp(-R_{\mathrm{scal}}\cdot R_i^2). \tag{15}$$

By definition, (9) describes an infinitely large volume. In [19], this volume is transformed into an ellipsoid-shaped volume by considering only the range within 3 times the standard deviations in each dimension. However, since this would cause all targets with the same standard deviations to occupy the same volume regardless of the RCSs, we use a threshold value $P_{\min}$ to limit $P_i^{(\mathrm{sc}_k)}$:

$$\begin{aligned} &P_i^{(\mathrm{sc}_k)}(R,\varphi,\theta) \\ &=\begin{cases} P_i^{(\mathrm{sc}_k)}(R,\varphi,\theta) & \forall P_i^{(\mathrm{sc}_k)}(R,\varphi,\theta)\geq\min(P_{\min},A_i) \\ 0 & \forall P_i^{(\mathrm{sc}_k)}(R,\varphi,\theta)<\min(P_{\min},A_i). \end{cases} \end{aligned} \tag{16}$$

$l\left(m_n^{(\mathrm{og})}(t)\,|act_{\mathrm{stat}}(t),\boldsymbol{X}_{\mathrm{ego}}(t)\right)$ is finally given the value $l_{\mathrm{occ}}\left(P_i^{(\mathrm{sc}_k)}\right)>0$ for each n affected by $P_i^{(\mathrm{sc}_k)}$.

3) NEGATIVE INVERSE SENSOR MODEL IN 3D

On the one hand, free spaces are assumed in the respective areas between sensor and target. Each of the areas $F_{1,i}^{(\mathrm{sc}_k)}$ represent elliptical cones with the opening angles $\boldsymbol{\Omega}_{1,i}=\left(\Omega_{\varphi,1,i},\Omega_{\theta,1,i}\right)^{\mathrm{T}}$ following from $P_i^{(\mathrm{sc}_k)}$ and assigns $l\left(m_n^{(\mathrm{og})}(t)\,|act_{\mathrm{stat}}(t),\boldsymbol{X}_{\mathrm{ego}}(t)\right)$ the constant value $l_{\mathrm{free},1}<0$ for each affected cell n. To incorporate the radar specificity that objects can be penetrated by radar waves, all those occupancy probabilities within $F_{1,i}^{(\mathrm{sc}_k)}$ that are also occupied by $P_j^{(\mathrm{sc}_k)}$, $i,j\in S_{\mathrm{act}}(t)$ will not be decreased by $l_{\mathrm{free},1}$. This is an essential difference to the common lidar ISMs.

On the other hand, we assume free space between sensor and a configurable distance R_{F_2} if no target has been detected in the field of view $FoV_{\varphi,k}$ of the k-th sensor in the azimuth dimension within an angle $\varphi_{\min}=\varphi_{\min}(v_{\mathrm{host},k})$. Here $\varphi_{\min}$ becomes smaller the higher the velocity $v_{\mathrm{host},k}$ of the k-th sensor, because with larger $v_{\mathrm{host},k}$ static targets at the same range have different velocities, so that an additional angular resolution is quasi achieved by the resolution in velocity dimension.

For each measurement cycle there are N_{F_2} such free space areas, which we denote with $F_{2,j}^{(\mathrm{sc}_k)}$ with $j=1,\ldots,N_{\mathrm{F}_2}$.

They have the corresponding opening angles $\boldsymbol{\Omega}_{2,j} = \left(\Omega_{\varphi,2,j}, \Omega_{\theta,2,j}\right)^{\mathrm{T}}$ and assign $l\left(m_n^{(\mathrm{og})}(t)\,|act_{\mathrm{stat}}(t), \boldsymbol{X}_{\mathrm{ego}}(t)\right)$ the value $l_{\mathrm{free},2} = l_{\mathrm{free},2}(R) < 0$ for each affected cell n. The value $l_{\mathrm{free},2}$ becomes larger with increasing range R. Furthermore, the beginning of the angle $\Omega_{\theta,2,j}$ is always chosen so that $N_{2,j}^{(\mathrm{sc}_k)}$ does not penetrate the ground plane level.

Finally, we use an area $F_3^{(\mathrm{gc})}$, which assigns exactly the logits $l\left(m_n^{(\mathrm{og})}(t)\,|act_{\mathrm{stat}}(t), \boldsymbol{X}_{\mathrm{ego}}(t)\right)$ to $l_{\mathrm{free},3} < 0$, which are occupied by the approximately cuboid volume of the test vehicle. Fig. 4 illustrates the individual areas of the ISM.

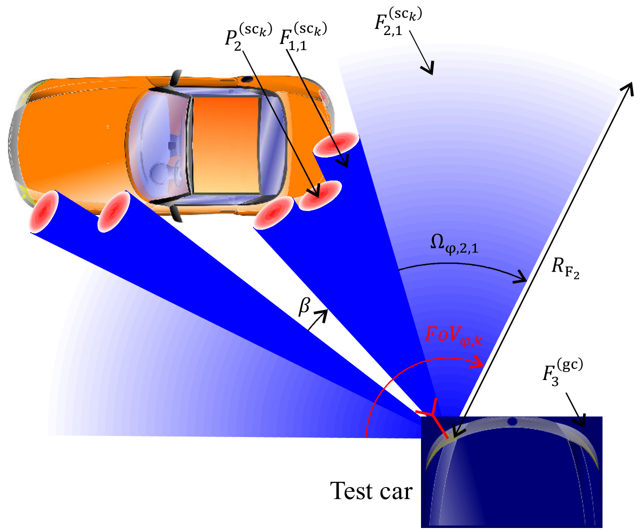

FIGURE 4. Representation of the areas of the ISM. The darker the coloring, the lower (blue) or higher (red) the probability of occupancy. The area $F_{2,1}^{(\mathrm{sc}_k)}$ is formed because $\Omega_{\varphi,2,1} \geq \varphi_{\mathrm{min}}$ is valid. However, in this example $\beta < \varphi_{\mathrm{min}}$.

4) IMPLEMENTATION IN 3D

The popular implementation variant OctoMap was presented in [20]. However, it is primarily intended for lidar data, so that SD_{φ} and SD_{θ} are approximated to zero. Thus, the OG processed with OctoMap based on radar data appears to be more sparsely occupied. Also, as this makes the filter process work worse, the OG has more clutter targets.

Therefore, we proceed similar to [19] by representing all $P_i^{(\mathrm{sc}_k)}$, $F_{1,i}^{(\mathrm{sc}_k)}$ and $F_{2,j}^{(\mathrm{sc}_k)}$ in the corresponding sensor coordinate systems sc_k in the discrete $R\varphi\theta$-space and then transferring them into the discrete, Cartesian xyz-space. This transfer is done by trilinear interpolation. The result is a temporal, Cartesian OG at measurement cycle t, compare Fig. 2 (d). The temporal OG is integrated into the global OG $\boldsymbol{m}^{(\mathrm{og})}(t)$ using (7) by leveraging our knowledge of the transformation matrices T_k and the test vehicle's current pose $\boldsymbol{X}_{\mathrm{ego}}(t)$.

Some parameters, such as l_{occ} and l_{free}, are difficult to dimension. They are therefore usually dimensioned empirically to allow the viewer to best associate the OG with the camera image. The authors of [21] showed that significant quantization errors occur for edge lengths $l_{\mathrm{og}} \geq 25$ cm. In this paper, we use $l_{\mathrm{og}} = 10$ cm. Fig. 5 demonstrates the difference between OctoMap and our implementation using an example.

5) SIMPLIFICATION TO 2D

The trilinear interpolation mentioned above is very computationally and memory intensive, which is why the use of a 2D representation is often favored. In our case, the 2D OG is a plane parallel to the xy-plane and considers only the targets from $act_{\mathrm{stat}}(t)$, for which $z_{\mathrm{min}} \leq z_{s_{\mathrm{act}}}^{(\mathrm{gc})} \leq z_{\mathrm{max}}$ applies. The interval $(z_{\mathrm{min}}, z_{\mathrm{max}})$ should be chosen such that the test vehicle is completely covered.

The 2D PDF to describe the probability of occupancy in the target's surroundings is

$$ {}^{(2\mathrm{D})}P_i^{(\mathrm{sc}_k)}(R,\varphi) = A_i \exp\left(-\left(\frac{(R-R_i)^2}{2SD_{\mathrm{R},i}^2} + \frac{(\varphi-\varphi_i)^2}{2SD_{\varphi,i}^2}\right)\right). \tag{17} $$

We use the same maximum value of the 3D PDF to ensure the comparability of both environment models. The 2D free spaces ${}^{(2\mathrm{D})}F_{1,i}^{(\mathrm{sc}_k)}$ and ${}^{(2\mathrm{D})}F_{2,j}^{(\mathrm{sc}_k)}$ are formed analogous to the 3D free spaces and correspond to circle sectors in the Cartesian coordinate system. Furthermore, the cuboid $F_3^{(\mathrm{gc})}$ becomes a rectangular surface.

In addition to the computational and storage overheads associated with the 3D implementation mentioned above, interpolation errors occur all the more during the transition from the discrete $R\varphi\theta$-space to the discrete xyz-space the smaller the distance to the sensor. We avoid this problem in the 2D implementation by rasterizing the affected areas of the ISM in the Cartesian xy-space using the Bresenham algorithm [22]. A 2D rasterization is much less computationally intensive than a 3D rasterization.

C. RCS GRID AND HIGHT GRID

Since the OG $\boldsymbol{m}^{(\mathrm{og})}$ has only information about the state of occupancy, we also use the RG $\boldsymbol{m}^{(\mathrm{rg})}$, which refers to the global coordinate system gc too. Analogous to the OG, $\boldsymbol{m}^{(\mathrm{rg})}$ contains an RCS $m_n^{(\mathrm{rg})}$ for each cell n with edge length l_{rg}, which characterizes material properties. Furthermore, the RCS is greatly dependent on the direction of arrival of the target, considering that very smooth vehicle parts reflect only a small amount of energy, while the very rough hubcaps are robust scattering centers. Since the test vehicle looks at a target from many directions as it drives past, we assign $m_n^{(\mathrm{rg})}(t)$ during accumulation the measured maximum to reduce the dependency on the direction of reception in RG:

$$ \begin{aligned} &\left(m_n^{(\mathrm{rg})}(t)\,|act_{\mathrm{stat}}(1\ldots t), \boldsymbol{X}_{\mathrm{ego}}(1\ldots t)\right) \\ &= \max\left(\left(m_n^{(\mathrm{rg})}(t)\,|act_{\mathrm{stat}}(t), \boldsymbol{X}_{\mathrm{ego}}(t)\right),\right. \\ &\qquad \left.\left(m_n^{(\mathrm{rg})}(t)\,|act_{\mathrm{stat}}(1\ldots t-1), \boldsymbol{X}_{\mathrm{ego}}(1\ldots t-1)\right)\right). \end{aligned} \tag{18} $$

Here, the first term in the max function describes the transformation of the current, static point cloud into a

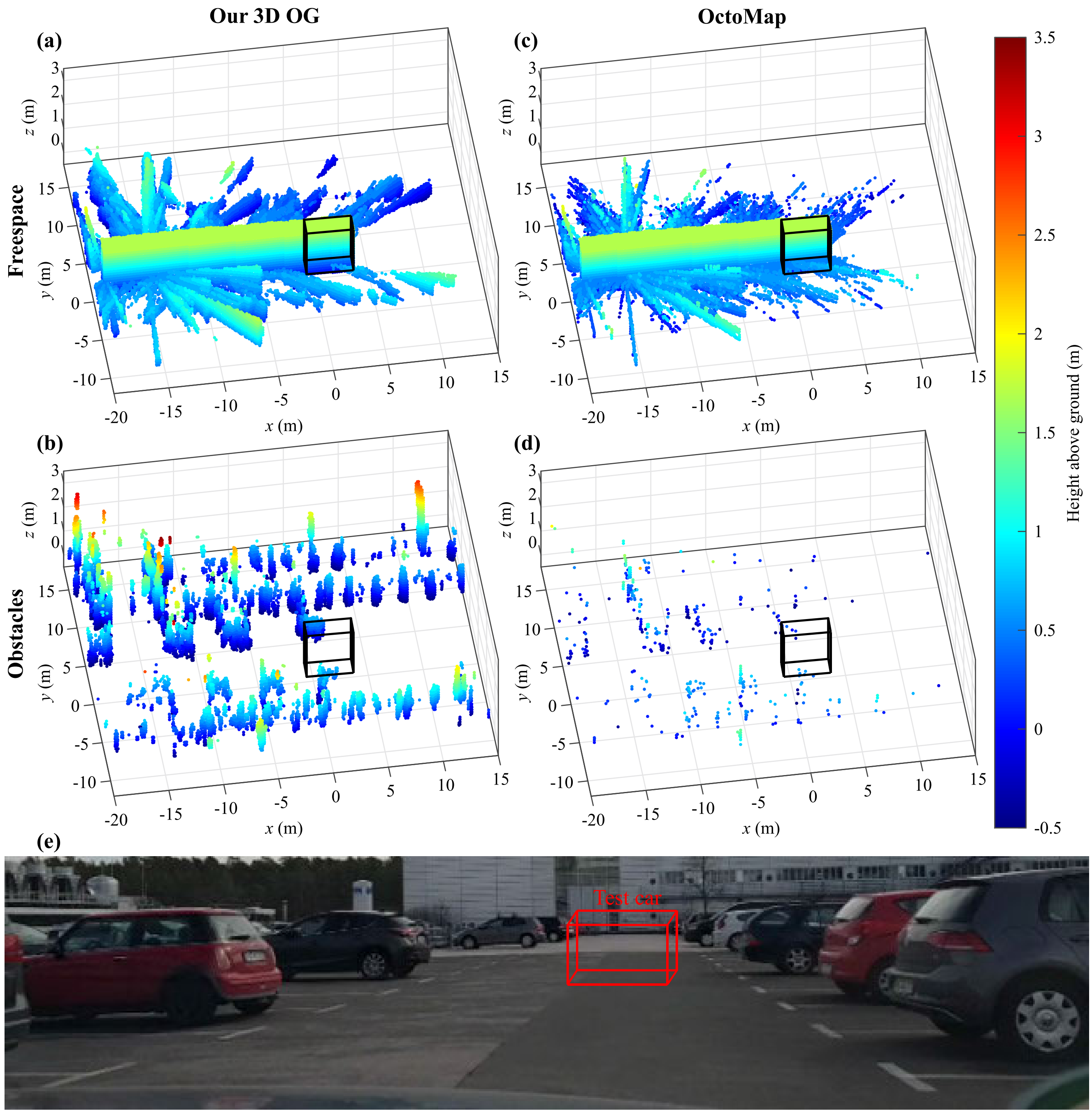

FIGURE 5. Comparison of the different 3D OG implementations. The red and black cuboids in (e) and (a)–(d), respectively, symbolize the test vehicle equipped with four corner radars. Only those voxels are shown that fall below a certain occupancy probability (free space in (a) and (c)) or exceed it (obstacles in (b) and (d)). The obstacles are so sparsely distributed in (d), because OctoMap occupies only one voxel for each $P_i^{(sc_k)}$.

current RG, which is then continuously integrated into $\boldsymbol{m}^{(\mathrm{rg})}$. A target always occupies exactly one cell. If multiple targets fall into the same cell, the maximum value is derived here as well. This procedure causes a quantization error that depends on the edge length. In this paper, we use $l_{\mathrm{rg}} = l_{\mathrm{og}} =$ 10 cm. Of course, this error is larger for a 2D implementation, because in this case significantly more targets fall into one pixel than into the individual voxels.

Finally, completely analogous to $\boldsymbol{m}^{(\mathrm{rg})}$ we use the height grid (HG) $\boldsymbol{m}^{(\mathrm{hg})}$, which represents the height $m_n^{(\mathrm{hg})}$ of the target above the ground for each cell with edge length $l_{\mathrm{hg}} = l_{\mathrm{rg}} = 10$ cm. The calculation is performed as in (18), with one difference that instead of the maximum value the arithmetic mean weighted with the reciprocal values of the measured ranges $R_{s_{\mathrm{act}}}$ is used to reduce the influence of far away targets. Figure 6 illustrates the difference between 2D and

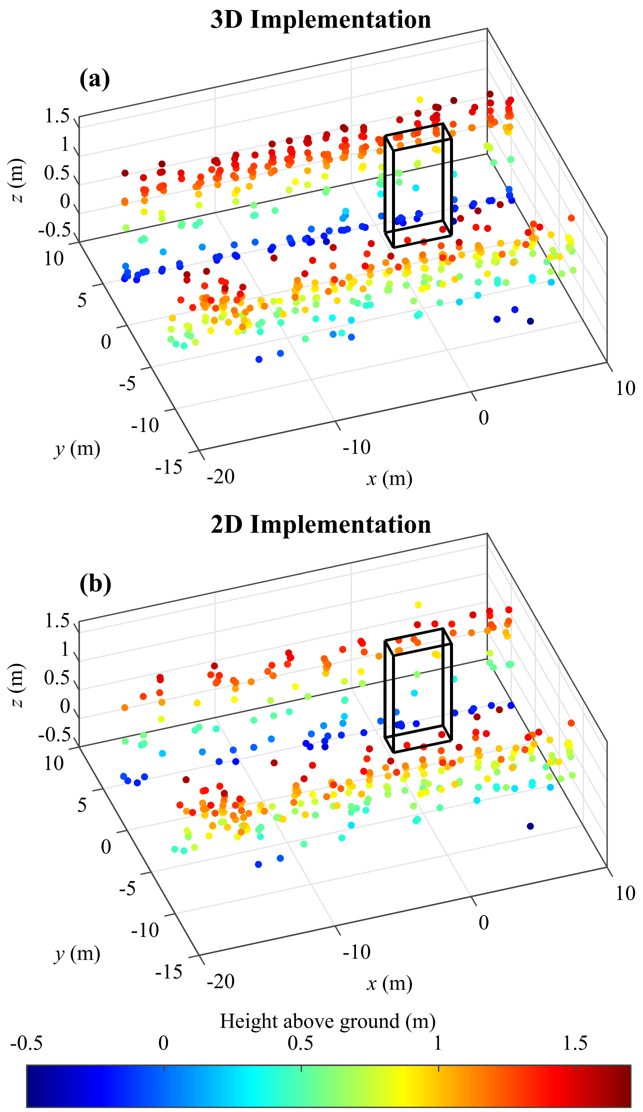

FIGURE 6. Comparison of 3D and 2D implementation of the HG for the scene in Fig. 1. While the targets in (a) are merged 3-dimensionally into the HG, the targets in (b) are merged only 2-dimensionally. Hence, the left guard rail appears to be more sparsely occupied in (b) than in (a).

3D implementation in terms of the height representation. Even with a 2D implementation – (18) is thus applied per xy-pixel and not per xyz-voxel – $\boldsymbol{m}^{(\mathrm{hg})}$ provides a well associable elevation profile.

III. NETWORK ARCHITECTURE

As already mentioned in Section I, we use CNN architectures to create SGs that perform image-to-image classification. Thus, each pixel in the input image is assigned to its corresponding class $c \in \{1, \ldots, C\}$. From the basic concept, such architectures follow an encoder-decoder structure, as shown in Fig. 7. While the encoder transforms the input image into a high-dimensional property vector by means of convolution and other operations, the decoder generates the output image from this property vector, the size of which corresponds to the input image. This requires so-called unpooling operations and/or transposed convolutions.

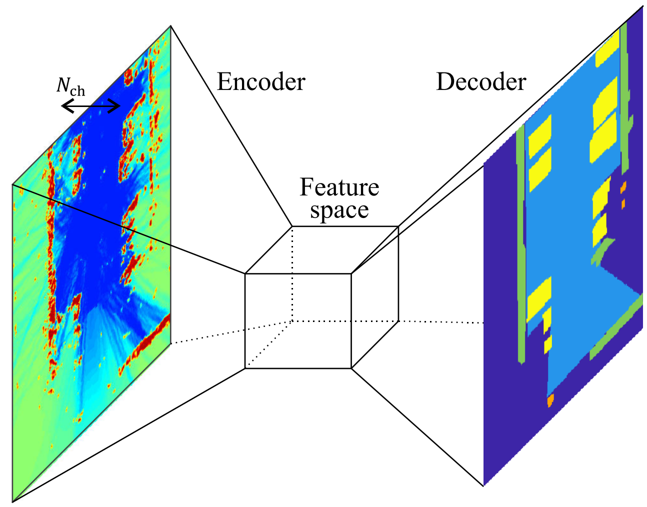

FIGURE 7. Schematic structure of the encoder-decoder architecture.

In [8], we used multiple CNNs – originally designed to segment camera images – for this purpose. The best results were achieved with SegNet-1 [23]. We also created SegNet-2, which consists of significantly fewer layers and thus requires less memory and fewer computing steps. We use ResNet-18-based DeepLabv3+ [24] as well in this paper: it performs significantly better than SegNet for camera image datasets.

The output layer of all three CNNs forms a softmax function that calculates the corresponding probability $p_c(\boldsymbol{x}_n)$ for each possible class c and pixel $\boldsymbol{x}_n$ in the output image. The probability $p_c(\boldsymbol{x}_n)$ in turn is the input variable of the cost function L, whose value is minimized during the training phase:

$$L = -\sum\nolimits_{\boldsymbol{x}_n} w_{\mathrm{bal}}\left(o\left(\boldsymbol{x}_n\right)\right) \log\left(p_{o(\boldsymbol{x}_n)}\left(\boldsymbol{x}_n\right)\right) + \frac{\lambda_{\mathrm{L2}}}{2} \cdot \boldsymbol{w}^{\mathrm{T}}\boldsymbol{w}. \tag{19}$$

Here, $o\left(\boldsymbol{x}_n\right)$ denotes the actual class from $\{1, \ldots, C\}$, $\boldsymbol{w}$ the vector of all weights in the CNN, and λ_{L2} the L2-regulation factor used to counteract overfitting. Furthermore, we use a weighting function $w_{\mathrm{bal}}(c)$, which weights a class the more strongly the smaller its pixel number $N_{\mathrm{pix}}(c)$ in the entire dataset:

$$w_{\mathrm{bal}}(c) = \sum\nolimits_{i=1}^{C} N_{\mathrm{pix}}(i) \big/ N_{\mathrm{pix}}(c). \tag{20}$$

As already mentioned in Section I, we use different types of input images, which are summarized in Table 1.

IV. DATASETS AND LABELING PROCEDURE

Since there is no suitable public dataset available at this time, we have created one simulation, and two real-world, datasets.

A. SIMULATION DATASET

To create the simulated dataset Γ_1 we used the Matlab Driving Scenario Designer (MDSD). With MDSD you can configure realistic scenarios and it provides you with scenario target lists. The dataset Γ_1 consists of parked cars, fences, and curbstones. The street is bordered almost continuously by

TABLE 1. **Input Images.**

Input	Number of input channels N_{ch}[a]	Description
E_1	3	$\boldsymbol{m}^{(\text{og})}$, $\boldsymbol{m}^{(\text{rg})}$, and $\boldsymbol{m}^{(\text{hg})}$ as 2D implementations
E_2	$\text{rnd}\left(\frac{z_{\max}-z_{\min}}{l_{\text{rg}}}\right)+2$	$\boldsymbol{m}^{(\text{og})}$ as 2D implementation, $\boldsymbol{m}^{(\text{rg})}$ as 3D implementation
E_3	$\text{rnd}\left(\frac{z_{\max}-z_{\min}}{l_{\text{og}}}\right)+\text{rnd}\left(\frac{z_{\max}-z_{\min}}{l_{\text{rg}}}\right)+2$	$\boldsymbol{m}^{(\text{og})}$ and $\boldsymbol{m}^{(\text{rg})}$ as 3D implementations

[a]$\text{rnd}(x)$ indicates the rounding result of x.

curbs or cars, which in most cases is what is encountered in the real world. The simulated test vehicle follows a set route at 30 to 50 km/h and is equipped with four corner radars whose range, resolution, accuracy and false alarm rate have been approximated to the performance of the short-range radars used for the Γ_2 dataset (see Section IV-B).

Dataset Γ_1 contains a total of 3040 images whose dimension are 384×384 pixels. This means that each image covers an area of 38.4×38.4 meters within the gc-related grid. The origin of the vehicle coordinate system is always in the center of the image, as shown in Fig. 8. In addition, each image is rotated so that the direction of view of the test vehicle is constant within all images. The images are rotated because the radar area coverages in the image are impacted only on the driven curve radius and therefore only vary slightly. Thus, the image quality in the lower part of the image is basically better and then becomes increasingly worse, because measurements in these upper areas were taken less frequently. Fig. 9 demonstrates the coverage areas assuming a typical radar radiation pattern. This outcome differs greatly from camera images whose quality remains almost constant over the entire image. In order to focus the CNN classification to the more frequently scanned areas, the coverage map from Fig. 9 can be integrated in (19).

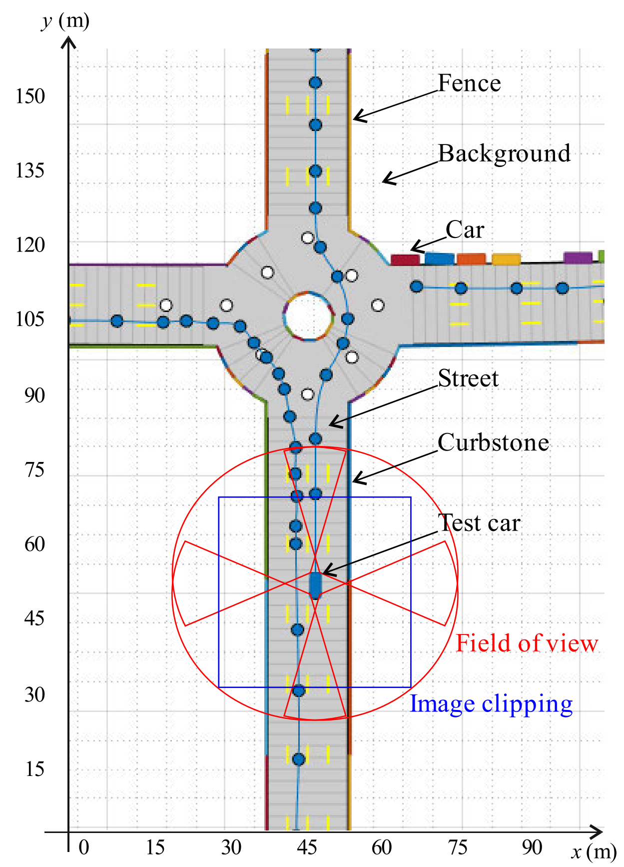

FIGURE 8. **Simulation environment. The test vehicle drives along the blue trajectory. The colors of the objects were arbitrarily chosen.**

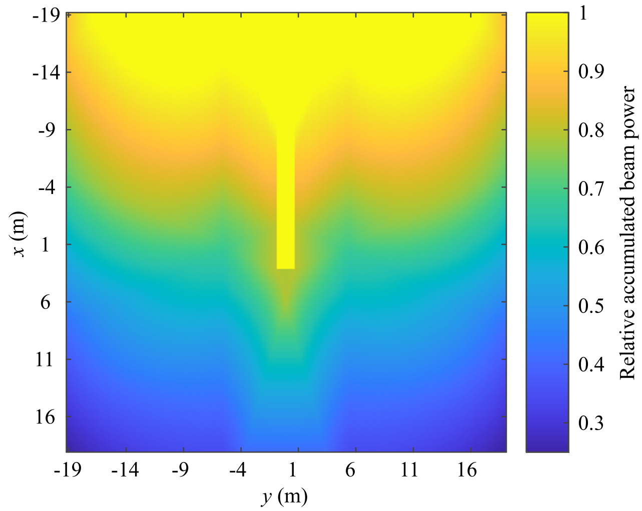

FIGURE 9. **Superposition of the radiation characteristics of four corner radars after a straight ride depicted in the vehicle coordinate system. A maximum value was assumed, which the calculated values do not exceed. The area traversed by the test vehicle was assigned this maximum value.**

In addition to the simulated objects, we also want to identify the street or the drivable underground, respectively. This region is largely calculated as free space in the OG, so that it is basically identifiable. This is different from [4], where only objects (obstacles) are classified. Because the size of the images to be classified is completely covered by the radar detection ranges except for small areas on the test vehicle, we use the simulated environment without any changes as the target output image of the segmentation. This means that sometimes areas without a measured radar target can be classified as objects like the undetected interior of a car, for example. However, since people can be trained to classify these types of areas, theoretically a CNN should be able to do so, as well.

B. REAL-WORLD I DATASET

Although simulations have the great advantage of being much more cost-saving than real-world measurements, there are also several limitations. In MDSD, for example, all objects are approximated in a cuboid shape, so that typical micro-Doppler signatures are no longer present and the object's RCS is characterized absolutely homogeneously along the surface. Furthermore, various object types are generally not supported, such as vegetation. As a consequence, the simulated environment has much less reflective structures than in reality, which makes it essential to perform real-world measurements to evaluate the method.

In a first test campaign, we equipped a test vehicle with four 79 GHz corner radars with respective ranges of 24 m and conducted measurements in typical parking scenarios, including covered car parks. The sensors used a bandwidth of 1.6 GHz and had a measurement accuracy of 1° and 2° in azimuth and elevation, respectively.

The campaign produced dataset Γ_2 containing 400 256 × 256 images. The image size has been reduced here because of the considerable amount of work involved in labeling in urban scenarios. Furthermore, the time between images is at least 3 s to keep the correlation between images low. Based on the measured scenes, we differentiate between the classes "Car", "Barrier", "Other" (obstacle), "Street" and "Background" in the source images.

Although the radar detection ranges in Γ_2 also cover the input images almost completely, objects are more often overlooked due to mirroring reflections in contrast to the simulation. Therefore, we do not use the ground truth data directly as the target output image of the segmentation here, but use the labeling result, which a trained person creates from the association between ground truth and OG. This is similar to labeling camera images. If the CNN succeeds in reproducing the segmentations created by the trained person, the SG obtained also classifies areas in which no target was detected due to the low resolution. This results in an additional gain in information, which is classically obtained using analytical techniques such as the Hough transformation. Fig. 10. illustrates the labeling procedure with an example.

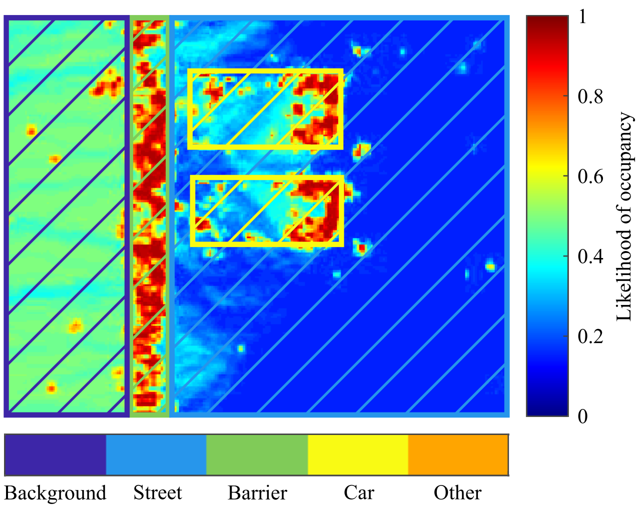

FIGURE 10. Labeling based on the OG. All pixels within the borders are assigned to the corresponding class.

C. REAL-WORLD II DATASET

Dataset Γ_3, consisting of 153 1536×512 large images, was created to assess the procedure as well. These comparatively large images could be labeled because this dataset was created on highways with a 76 GHz front radar, whose bandwidth was 1 GHz and the maximum range 300 m. The measurement accuracy of azimuth and elevation was 0.15° each. Only the barrier, road and background classes were distinguished for this dataset because of the lower scenario complexity. The labeling procedure itself was the same as for Γ_2.

Table 2 summarizes the most important characteristics of the three datasets.

TABLE 2. Datasets.

Dataset	Images	Image size	Classes
Γ_1 (4 corner radars)	3040	384×384	Background, Street, Car, (high) Fence, Curbstone
Γ_2 (4 corner radars)	400	256×256	Background, Street, Car, Barrier, Other (obstacle)
Γ_3 (1 front radar)	153	1536×512	Background, Street, Barrier

V. EXPERIMENTS AND RESULTS

Each of the three datasets was split into 80% training data and 20% test data. For synthetic enhancement, all images were mirrored along the x-axis of the vehicle coordinate system. Since the datasets are still relatively small, we also performed a three-fold cross-validation, resulting in three different splits in training and test data for each dataset.

The training was performed with the stochastic gradient descent with momentum algorithm for 50 epochs each. The initial learning rate was 0.01 and was halved every ten epochs. Momentum and L2 regularization factor were set to 0.9 and 5e-4, respectively. Since the 3D input images are very memory intensive, the batch size was set to 1. The calculation was done on a Linux system with an installed Nvidia Quadro GV100 graphics card.

Basically, SGs offer the advantage over other grids in that they incorporate a human interpretation of the radar data. A few radar targets of an object are automatically assembled to a characteristic shape and then assigned to an object class. Although these shapes usually have a coarser structure with greater spatial inaccuracies, this fact is not of great importance since the radar point cloud is already present without any rounding errors. The combination of SG and point cloud results in an improved scene understanding.

All results are summarized in Tables 3 to 5, with the best values highlighted in bold. The parameters Recall, Jaccard coefficient (IoU) and F1 contour match score (BF), which are common in this context, were calculated and averaged across the respective classes.

Despite the "stricter" labeling (cf. Section IV-A), the Γ_1 results clearly exceed the Γ_2 results, which are similar in terms of the scenarios, because the simulation data a) provide

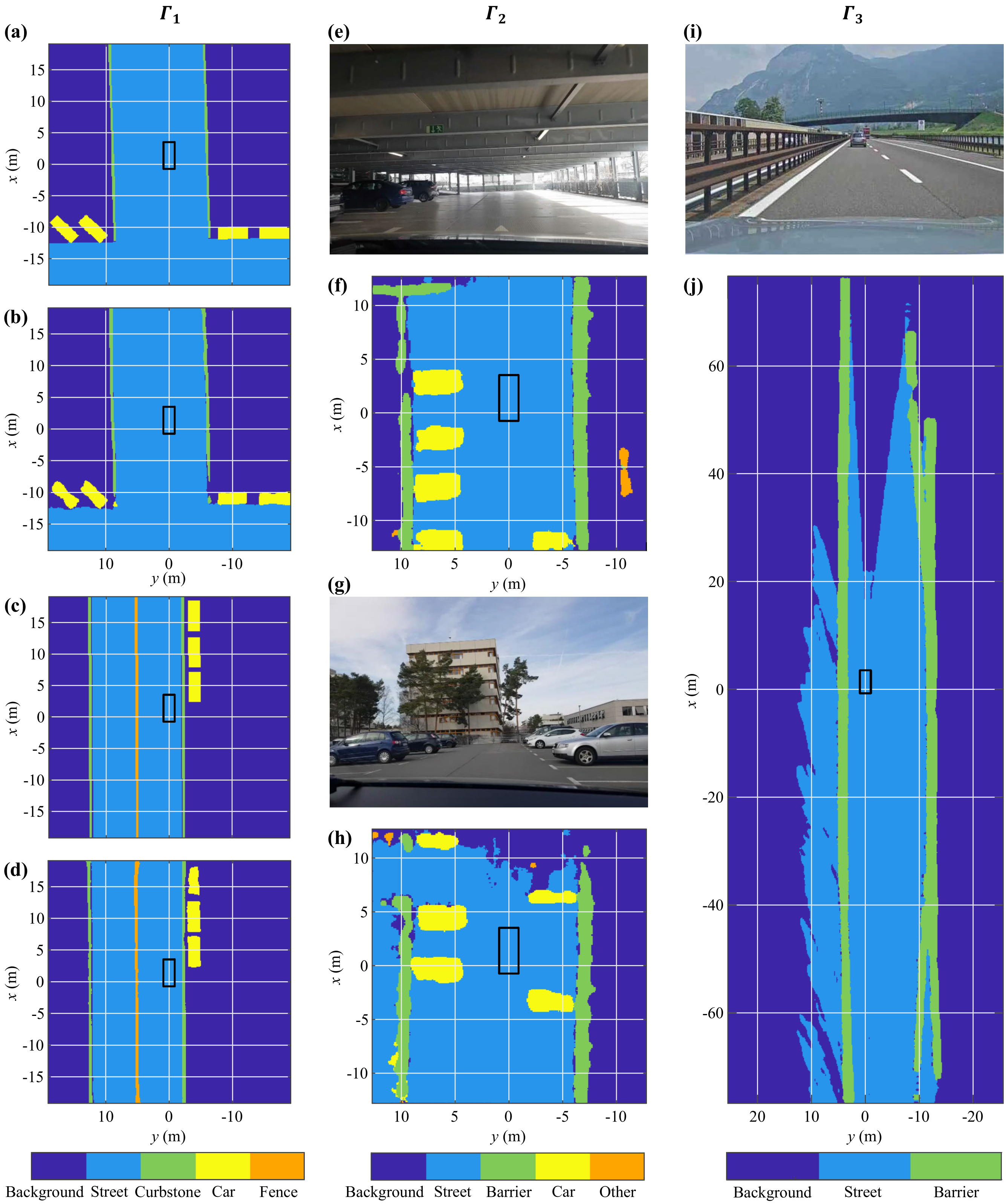

FIGURE 11. **Comparison of ground truth and SG. For Γ_2 and Γ_3, the image pairs (e) and (f), (g) and (h) as well as (i) and (j) are not time-synchronous, since the photographs are intended to provide an overview of the scene already passed by the test vehicle.**

a faultless trajectory of the test vehicle and b) follow a much more limited pattern with regard to the radar targets than the real-world data. Thus, in the real-world data, especially in parking garages, systematic, strong clutter centers occur, which result from multipaths and are generally not modeled in the MDSD. This leads mainly to differences in the

TABLE 3. Results Γ_1.

Input	Network	Recall	IoU	BF
E_1	SegNet-1	0.954	0.838	0.962
	SegNet-2	0.817	0.666	0.798
	DeepLabv3+	**0.980**	0.839	**0.974**
E_2	SegNet-1	0.955	0.843	0.970
	SegNet-2	0.827	0.681	0.813
	DeepLabv3+	0.977	**0.849**	0.973
E_3	SegNet-1	0.952	0.844	0.971
	SegNet-2	0.837	0.687	0.818
	DeepLabv3+	0.974	**0.849**	0.970

TABLE 4. Results Γ_2.

Input	Network	Recall	IoU	BF
E_1	SegNet-1	0.904	0.773	0.810
	SegNet-2	0.898	0.701	0.710
	DeepLabv3+	0.866	0.746	0.763
E_1[a]	SegNet-1	0.908	0.744	0.800
E_2	SegNet-1	0.906	0.802	0.824
	SegNet-2	0.900	0.724	0.718
	DeepLabv3+	0.906	0.780	0.793
E_3	SegNet-1	**0.916**	**0.812**	**0.829**
	SegNet-2	0.902	0.733	0.725
	DeepLabv3+	0.909	0.800	0.801

[a]This result was achieved following the example in [4]: No OG cell falls below the probability 0.5 and the classes background and street are combined.

TABLE 5. Results Γ_3.

Input	Network	Recall	IoU	BF
E_1	SegNet-1	0.959	0.850	0.928
	SegNet-2	0.953	0.835	0.903
	DeepLabv3+	0.946	0.876	**0.962**
E_2	SegNet-1	0.961	0.842	0.933
	SegNet-2	0.954	0.845	0.913
	DeepLabv3+	0.944	**0.877**	0.961
E_3	SegNet-1	**0.962**	0.853	0.930
	SegNet-2	0.953	0.840	0.909
	DeepLabv3+	0.947	0.872	0.959

background class. Furthermore, in reality, cars in particular split up into dominant clutter centers around the tires, whereas the car body often reflects only mirroring at the doors. Consequently, cars in the corresponding grids show different appearances depending on the situation, while in the simulation the concise L- or U-shape is mostly used.

The Γ_3 results on the other hand are not worse than the Γ_1 results. This is mainly due to the fact that there is only one object class in this dataset and the network therefore only has to learn to distinguish barriers that occur as a systematic sequence of radar targets from individual targets of the background class.

Common to all three datasets is that SegNet-1 and DeepLabv3+ achieve comparable results that are significantly better than those of SegNet-2. This was to be expected due to the strong reduction of the network architecture. Nevertheless, SegNet-2 also delivers useful results, which should be emphasized in view of the low memory requirement of the network of only about 3 MB.

With regard to the benefits of using 3D grids, the picture is divided. There are no significant differences between datasets Γ_1 and Γ_3. This can be explained by the fact that a) in the simulation the height information $\boldsymbol{m}^{(\mathrm{hg})}$ is already sufficient due to the small variation of the patterns in the grids and b) in Γ_3 only one object class exists. Regarding Γ_2, however, slightly better results are obtained for the inputs E_2 and E_3. However, it should be noted that the trilinear interpolation in the 3D implementation of $\boldsymbol{m}^{(\mathrm{og})}$ results in range and cell size dependent interpolation errors that do not occur in the 2D implementation in this form. Thus, the near range of the 3D implementation appears to be of lower quality.

As already mentioned in Section IV, in [4] the free spaces in the OG are reset and thus no distinction is made between the street and the background in the area free of obstacles. In Table 4, a comparison shows that the omission of the free space information tends to have a negative effect on the classification of the objects. This seems plausible in view of the special free space patterns around the objects.

Fig. 11 demonstrates different segmentation results generated by SegNet-1 for the input E_3. The good results from Table 2 are also recognizable in the SGs (b) and (d), which are very similar to the simulated environments (a) and (c). Regarding Γ_2, the quality of the SGs is noticeably worse. Besides minor confusions between cars and barriers, especially the contours of the cars in (f) and (h) deviate from ground truth. In the SG (j) the barriers appear thicker than the real guardrails because the standard deviation SD_φ was chosen relatively large due to the small number of accumulated targets caused by the high speed of the test vehicle. The high speed is also the reason for the worse road detection straight ahead of the test vehicle, because the free space F_2 is measured more rarely in this case.

VI. CONCLUSION

As a result of our investigations we can state that 3D input images do not lead to significant advantages for the simulation dataset as well as the dataset with simple infrastructure. The height information of the 2D HG seems to be sufficient here. Regarding the dataset with more complex infrastructure, however, the results are slightly better for the 3D input. Furthermore, we can see that the use of free space information in the OGs acting as input images is advantageous. In terms of the influence of the used network, it can be stated that useful results are already achieved with a significantly reduced SegNet variant. Surprisingly, the DeepLabv3+ in our radar datasets, which is very performant in camera images, leads to at most minor advantages over the original SegNet.

A larger dataset is required to further analyze and optimize the technique presented. The future focus will be on automated labeling algorithms, as manual labeling is only supported by vehicle-mounted cameras and is very labor intensive. Here, the use of other reference measurement technology is conceivable, such as vehicle lidars and 360° camera systems.

Automotive radars that measure 3D are currently being developed. In particular, the resolution in elevation will improve significantly over the coming years. Using such sensors, 3D segmentation is useful for scenes with particularly complex infrastructure, so that the output image classifies not only objects close to the ground but also roofs, for example – a key element for autonomous vehicle guidance in indoor scenarios. For this problem, the use of 3D input images would be indispensable.

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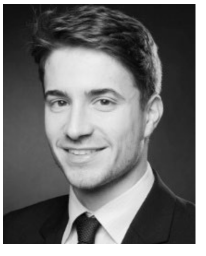
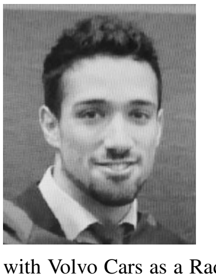


ROBERT PROPHET (Student Member, IEEE) received the bachelor's and master's degrees in electrical engineering from the University of Applied Sciences Leipzig, Leipzig, Germany, in 2012 and 2015, respectively. In 2015, he joined the Institute of Microwaves and Photonics, Friedrich-Alexander-University Erlangen-Nuremberg, Erlangen, Germany. His research interests include automotive radar signal processing and machine learning techniques.



ANASTASIOS DELIGIANNIS (Member, IEEE) received the diploma degree (bachelor's and master's degrees equivalent) from the School of Electrical and Computer Engineering, University of Patras, Patras, Greece, in 2012, and the Ph.D. degree in radar signal processing from Loughborough University, Loughborough, U.K., in 2016. He remained with Loughborough University until 2018 as a Postdoctoral Research Associate of signal processing. From 2018 to 2019, he worked with Volvo Cars as a Radar Performance Engineer for autonomous driving projects. Since July 2019, he has been with the BMW Group as a Radar Expert of autonomous driving and driving assistance projects. His research interests include signal processing algorithms, sparse array design, convex optimization, beamformer development, and game theoretic methods, within the radar network framework and wireless communications.

JUAN-CARLOS FUENTES-MICHEL received the Ph.D. degree from the Clausthal University of Technology, Germany, in 2008. He is currently a Radar Technology Expert for driver assistance systems and autonomous driving with the BMW Group and the author of several publications on wireless positioning and radar signal processing. During his career, he has contributed to the industrialization of different radar components in the automotive industry, from the initial concept phase until mass implementation.

INGO WEBER received the master's degree in mechanical engineering from Cornell University, in 1999, and the diploma and Ph.D. degrees in vehicle technology and feedback control systems from the Technical University of Darmstadt, in 2000 and 2004, respectively. He got the Carl-Schenk-Award in 1999. In 2003, he started to work at BMW and was part of the development of dynamic stability control systems (DSC) and of integrated chassis management (ICM). He was the Head of the test and validation team from 2008 to 2010 and the Module Lead for chassis control and driver assistance systems in numerous car projects from 2011 to 2015. Since the beginning of 2015, he has been the Head of radar development with the BMW Group and develops automated driving with his radar group. He works with his team and bachelor's, master's, and Ph.D. students on new radar hardware and new radar algorithms to prepare the radar technology for the next levels of automated driving.

MARTIN VOSSIEK (Fellow, IEEE) received the Ph.D. degree from Ruhr-Universität Bochum, Bochum, Germany, in 1996. In 1996, he joined Siemens Corporate Technology, Munich, Germany, where he was the Head of the Microwave Systems Group, from 2000 to 2003. Since 2003, he has been a Full Professor with Clausthal University, Clausthal-Zellerfeld, Germany. Since 2011, he has also been the Chair of the Institute of Microwaves and Photonics (LHFT), Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Erlangen, Germany. He has authored or coauthored more than 250 articles. His research has led to over 90 granted patents. His current research interests include radar, transponder, RF identification, communication, and locating systems. He has been a member of organizing committees and technical program committees for many international conferences. He is also a member of the German IEEE Microwave Theory and Techniques (MTT)/Antennas and Propagation (AP) Chapter Executive Board and the IEEE MTT-S Technical Coordinating Committees MTT-24, MTT-27, and MTT 29. He was a recipient of several international awards. For example, recently, he was awarded the 2019 Microwave Application Award from the IEEE MTT Society (MTT-S). He was the Founding Chair of the MTT IEEE Technical Coordinating Subcommittee MTT-27 Wireless-Enabled Automotive and Vehicular Application. He has served on the review boards for numerous technical journals. From 2013 to 2019, he was an Associate Editor of the IEEE Transactions on Microwave Theory and Techniques (MTT).